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Chromatic Scales?

Question

Do musical keys have audible personalities, and can we see them?

Spotify assigns every track a key (0–11, C through B) along the chromatic scale. A chromatic scale consists of all 12 notes in an octave.

I wanted to lay out the 12 Chromatic keys in their natural physical order and ask whether the music people stream in each key differ on Spotify's seven audio features.

For each key I computed the mean of every feature across all streams, then min-max rescaled so the lowest key maps to 0 and the highest to 1. Significant deviations are identified with a robust z-score ($\text{median} + \text{MAD} \times 1.4826$); any key whose top feature exceeds $z = 1$ earns a personality label.

Visual Strategy

The final visualization, *Chromatic Scales?*, uses three spatial structures working together.

Piano keyboard as index. A one-octave keyboard (C4–B4) spans the top of the chart, colored by personality. It anchors the reader in the chromatic ordering and doubles as a clickable filter in the interactive version.

Radar small multiples in chromatic order. Below the keyboard, each key gets a card with a 7-axis radar. Cards run left to right, C through B, matching the keyboard above. Each radar overlays two polygons: a dashed grey median (the benchmark across all 12 keys) and a colored data polygon for that key. Where the colored shape exceeds the grey one, the key is above median. Significant deviations ($|z| \geq 1$) get filled dots; non-significant points are dimmed. The viewer reads shape difference, not axis values.

Spotlight and annotation. Keys with a distinctive personality get a bold colored border and in-cell callout text explaining what stands out and naming a representative song. Keys that fall within the normal range are dimmed to 35% opacity. This foreground/background treatment lets the eye skip to the interesting keys without losing the full context of all 12.

A glossary table at the bottom defines each personality label so the chart is self-contained.

Feedback and Iteration

After the first draft, I received seven pieces of feedback that reshaped the final version.

1. Remove the +/- significance tags.

The first draft placed small colored pills on each radar axis showing whether a feature was significantly above or below the median. In a static export these were hard to read and cluttered the radars. I removed them entirely; the filled-dot versus dimmed-dot encoding carries the same information more quietly.

2. More “why”: do top songs demonstrate the groups?

The original chart showed personalities but gave no evidence. The final version adds inline callout annotations beneath each distinctive key, naming the top-streamed song in that key and explaining, in one sentence, why the personality fits. For example, C#/D $\flat$ (“The Groover”) cites “Or Nah” by Ty Dolla \$ign; A#/B $\flat$ (“The Storyteller”) cites “Ten Duel Commandments” from *Hamilton*.

3. Lay the facets in keyboard order.

The first draft grouped cards by personality, which meant related keys were next to each other, but the chromatic sequence was lost. The final chart runs strictly C through B, left to right, matching the piano above. This makes it easy to scan across the octave and notice that sharp/flat keys tend to cluster differently from naturals.

4. Title as a question.

The first draft used a declarative title. The final version asks, “*Chromatic Scales?*” posing the question the graphic answers.

5. Do these really represent personalities?

This was the hardest feedback. The personality labels are a shorthand for “the feature with the highest z-score above 1.” They are descriptive, not causal. I kept the labels but leaned into the evidence: each callout ties the label to a concrete song and a plain-English explanation. The glossary defines the threshold so a skeptical reader can evaluate the claim.

6. Left-to-right readability.

Addressed by the switch to chromatic ordering (point 3) and the spotlight/dim treatment, which creates a visual rhythm across the row: bold, dim, bold, dim.

7. Post-production refinement.

To play with formatting the svg more in post-production.

So What?

Musical keys are not neutral containers. The chart shows that 8 of 12 keys have at least one feature scoring more than 1σ above the median:

- **Sharp/flat keys lean instrumental.**
D, F#/Gb, and A all earn “The Silent Type,” reflecting that ambient, lo-fi, and classical music gravitates to these keys.
- **C#/Db is the groove key.**
It has the highest rescaled danceability, with steady-beat tracks dominating its top streams.
- **E and B sound live.**
“The Stage Lover” label appears for keys heavy in concert recordings and audience energy.
- **A#/Bb is word-dense.**
Rap, spoken word, and storytelling tracks cluster here.
- **D#/Eb is the happiest key.**
It scores highest on valence.

The four remaining keys (C, F, G, G#/Ab) sit near the median on everything, hosting the broadest mix of content with no dominant personality.

Key choice is not independent of musical style. Composers and producers write certain kinds of music in certain keys, for physical reasons (guitar tunings favor E, A, G; piano pieces sit in C, F, Bb) or cultural convention. This visualization makes that latent structure visible.

What I Would Change

1. **Weight by listening time.** Currently each stream counts equally. Weighting by `msPlayed` would better reflect what listeners heard versus what they skipped.
2. **Split by mode.** The chart ignores major versus minor. Splitting into 24 categories (12 keys × 2 modes) might reveal that the major/minor distinction drives personality differences more than pitch class does.
3. **Tighter annotation density.** Some callouts are long for a static export. A next pass would trim each to a single line and move longer explanations to a companion piece.